We assume the existence of the sets $\mathbb{N} = \{1, 2, 3, \dots\}$ of natural numbers, $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ of integers, $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ of rationals, and $\mathbb{R}$ of all reals. We assume that $\mathbb{R}$ has the operations of *addition* (denoted $a + b$) and *multiplication* (denoted ab or $a \cdot b$) which satisfy the following.

Field axioms: Properties of Addition and Multiplication

1. **Closure:** If $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.
2. **Commutativity:** If $a, b \in \mathbb{R}$, then $a + b = b + a$ and $ab = ba$.
3. **Associativity:** If $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
4. **Distributivity:** If $a, b, c \in \mathbb{R}$, then $(a + b)c = ac + bc$ and $a(b + c) = ab + ac$.
5. **Existence of Identities:** There are two different real numbers 0 and 1 such that for all real numbers a, $a + 0 = a$ and $a \cdot 1 = a$.
6. **Additive Inverses:** For all $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
7. **Multiplicative Inverses:** For all $a \in \mathbb{R}$ such that $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

You are also allowed to assume: $\mathbb{N} \cup \{0\}$ satisfies Axioms 1–5 (where the closure property needs to be restated for $\mathbb{N}$, and all the quantifiers should be quantified over $\mathbb{N}$), $\mathbb{Z}$ satisfies Axioms 1-6 (again where closure means for $\mathbb{Z}$ and quantifiers are over $\mathbb{Z}$), and $\mathbb{Q}$ also satisfies Axioms 1–7. This is because the real numbers are not the only field.

Equality Axioms

These hold for $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$. We state them for $\mathbb{R}$.

8. **Reflexivity:** For every real number a, $a = a$.
9. **Symmetry:** For all real numbers a and b, if $a = b$, then $b = a$.
10. **Transitivity:** For all real numbers a, b, and c, if $a = b$ and $b = c$ then $a = c$.
11. **Substitution:** For all real numbers a and b, if $a = b$, then b may be substituted for a in any number of places within a mathematical statement without changing the truth value of that statement.

Order Axioms

We assume that $\mathbb{R}$ contains a subset $\mathbb{R}^+$, whose elements are called "positive real numbers", and that satisfies the following properties

12. **Positive Closure:** If a and b are positive real numbers, then $a + b$ and ab are positive.
13. **Trichotomy:** If a is a real number, then exactly one of the following is true:
    - a is positive,
    - $-a$ is positive,
    - $a = 0$.

We define the set positive integers and positive rationals by $\mathbb{Z}^+ = \mathbb{Z} \cap \mathbb{R}^+$ and $\mathbb{Q}^+ = \mathbb{Q} \cap \mathbb{R}^+$, respectively. These also satisfy the properties of positive closure and trichotomy.

The last axiom only holds for sets of *positive integers*.

14. **Well-Ordering Property:** If S is a nonempty set of positive integers, then S has a minimum element.